

MOVING CHARGES AND MAGNETISM

Chapter Four
Class - XII

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Learning Objectives:

- Concept of magnetic field, Oersted's experiment.
- Biot - Savart law and its application to current carrying circular loop.
- Ampere's law and its applications to infinitely long straight wire. Straight solenoid (only qualitative treatment).
- Force on a moving charge in uniform magnetic and electric fields.
- Force on a current-carrying conductor in a uniform magnetic field, between two parallel current-carrying conductors-definition of ampere,

Oersted's Experiment

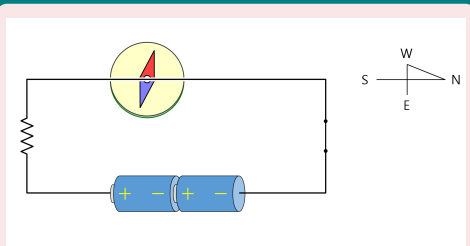


Figure: Oersted's Experiment Setup.
Source : Java Lab

- Danish physicist **Hans Christian Oersted** **accidentally discovered** that a **current in a straight wire** caused a noticeable **deflection** in a nearby **magnetic compass needle**.
- **Oersted experimented** with this phenomenon several times, **revealing a close relationship between electrical and magnetic phenomena**.

Oersted's Experiment

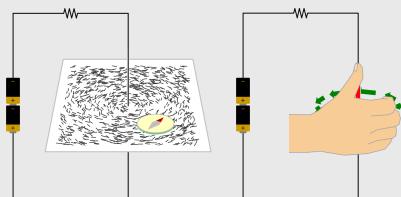


Figure: Magnetic Field around a Wire.
Source : Java Lab

Observations and Results:

- He found that the **alignment of the needle is tangential to an imaginary circle** which has the straight wire as its center and has its plane perpendicular to the wire.
- **Reversing** the direction of the **current** **reverses the orientation of the needle.**
- The **deflection increases on increasing the current** or **bringing the needle closer to the wire.**

Oersted's Experiment

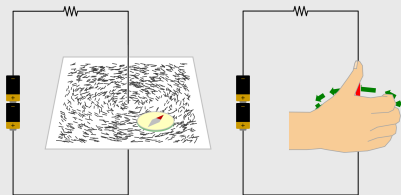
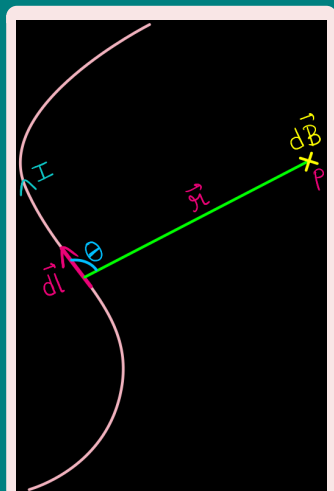


Figure: Magnetic Field around a Wire and its direction is give by right hand thumb rule. Source : Java Lab

Observations and Results:

- **Iron filings** sprinkled around the wire **arrange themselves in concentric circles** with the wire as the center.
- The **magnetic field** is the **area surrounding a magnet** in which the **magnetic force exists**.
- **Oersted concluded** that **moving charges or currents produced a magnetic field** in the surrounding space.

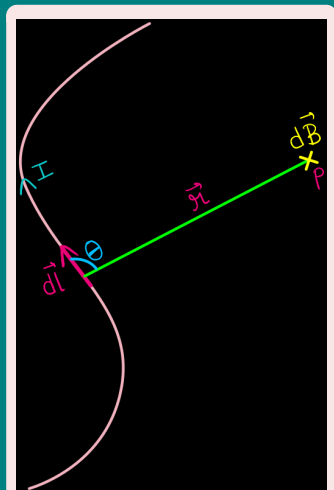
BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- Figure shows a finite conductor carrying **current I** .
- Consider an **infinitesimal element $d\vec{l}$** of the conductor.
- The **magnetic field $d\vec{B}$** due to this element is to be determined **at a point P** which is at a **distance r** from it. Let θ be the **angle between $d\vec{l}$ and the displacement vector r** .

BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- According to Biot-Savart's law, the magnitude of the magnetic field $d\vec{B}$ is proportional to

- 1 The current I ,

$$(d\vec{B} \propto I)$$

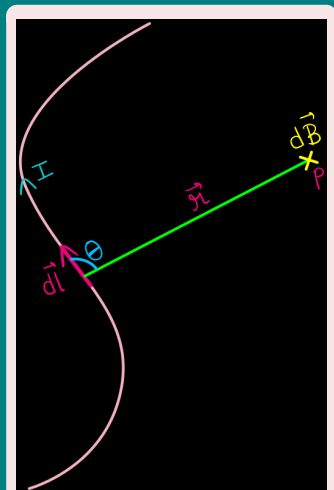
- 2 The element length dl ,

$$(d\vec{B} \propto dl) \quad \text{and}$$

- 3 Inversely proportional to the square of the distance r

$$\left(d\vec{B} \propto \frac{1}{r^2} \right).$$

BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- **direction** of $d\vec{B}$ is **perpendicular to the plane** containing $d\vec{l}$ and $\vec{r}$. Thus, in vector notation,

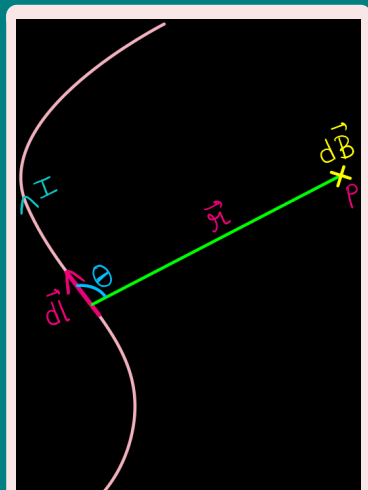
$$d\vec{B} \propto \frac{I d\vec{l} \times \hat{r}}{r^2}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

or

$$|d\vec{B}| = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

BIOT-SAVART LAW



Magnetic Field Due to a Current Element

- where $\frac{\mu_0}{4\pi}$ is a **constant of proportionality** in SI units has the exact value.

$$\frac{\mu_0}{4\pi} = 10^{-7} \text{ Tm/A}$$

- We call μ_0 the **permeability of free space (or vacuum)**.

Similarities and Differences b/w Biot-Savart and Coulomb's law

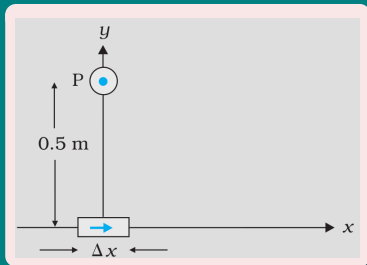
- **Both are long range**, since **both depend inversely** on the **square of distance** from the source to the point of interest.
- The **principle of superposition applies to both** fields.
- The **electrostatic field is produced by a scalar source**, namely, the electric charge. The **magnetic field is produced by a vector source** $I d\vec{l}$.
- The **electrostatic field is along the displacement vector joining the source and the field point**. The **magnetic field is perpendicular to the plane containing the displacement vector $\vec{r}$ and the current element $I d\vec{l}$** .
- There is an **angle dependence in the Biot-Savart law** which is **not present in the electrostatic case**.

Relation between Permittivity of free space (ϵ_0) and the permeability of free space (μ_0)

- There is an **interesting relation** between ϵ_0 , the **permittivity of free space**; μ_0 , the **permeability of free space**; and c , the **speed of light in vacuum**:

$$\begin{aligned}\epsilon_0 \times \mu_0 &= 4\pi\epsilon_0 \times \frac{\mu_0}{4\pi} \\ \epsilon_0 \times \mu_0 &= \frac{1}{9 \times 10^9} \times 10^{-7} \\ \epsilon_0 \times \mu_0 &= \frac{1}{9 \times 10^{16}} = \frac{1}{(3 \times 10^8)^2} \\ \epsilon_0 \times \mu_0 &= \frac{1}{c^2} \quad \boxed{c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}}\end{aligned}$$

Example 4.5



An element $\Delta l = \Delta x \hat{i}$ is placed at the origin and carries a large current $I = 10 \text{ A}$. What is the magnetic field on the y-axis at a distance of 0.5 m. $\Delta x = 1 \text{ cm}$.

Applications of Biot-Savart Law

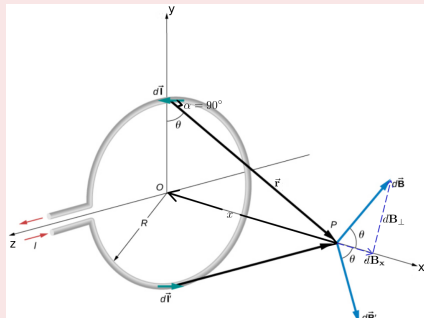


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- Figure depicts a **circular loop** carrying a **steady current I** .
- The **loop** is placed in the **y - z plane** with its center at the **origin O** and has a **radius R** .
- **We wish to calculate the magnetic field at the point P** on this axis.
- Let **x** be the **distance of P** from the center **O** of the loop.

Applications of Biot-Savart Law

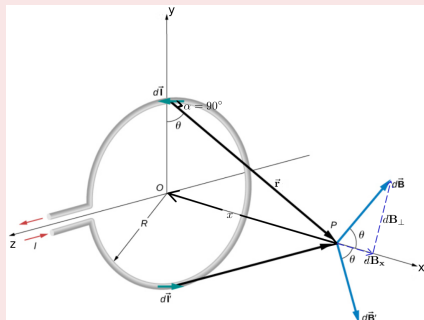


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- The magnitude dB of the magnetic field due to $d\vec{l}$ is given by the Biot-Savart law

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \alpha}{r^2}$$
$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \alpha}{(x^2 + R^2)} \quad (\because r^2 = R^2 + x^2)$$

- The element $d\vec{l}$ will be perpendicular to the displacement vector $\vec{r}$ (Or $\alpha = 90^\circ$)

Applications of Biot-Savart Law

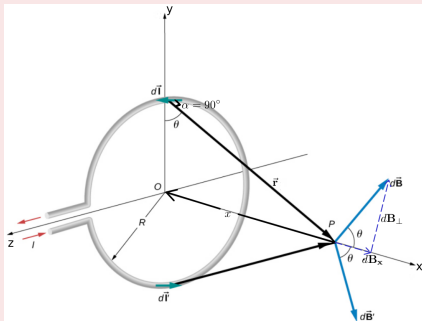


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- Thus,

$$dB = \frac{\mu_0}{4\pi} \frac{Idl}{(x^2 + R^2)^{3/2}} \quad \dots (1)$$

- $d\vec{B}$ has an **x-component** dB_x and a **component perpendicular** to x-axis, $dB_{\perp}$.
- When the **perpendicular components** ($dB_{\perp}$) are **summed over**, they **cancel out**.
- Thus, only the x-component (dB_x) survives

Applications of Biot-Savart Law

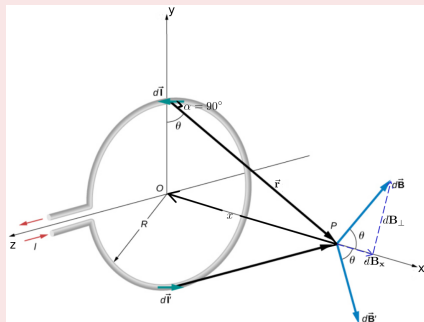


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

Magnetic Field On The Axis Of a Circular Current Loop

- The **net contribution along x-direction** can be obtained by integrating $dB_x = dB \cos \theta$ over the loop.

$$\cos \theta = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}} \dots\dots(2)$$

- From Eqs. (1) and (2),

$$dB_x = \frac{\mu_0}{4\pi} \frac{Idl}{(x^2 + R^2)} \times \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

Applications of Biot-Savart Law

Magnetic Field On The Axis Of a Circular Current Loop

- $$dB_x = \frac{\mu_0}{4\pi} \frac{IRdl}{(x^2 + R^2)^{\frac{3}{2}}}$$

- Thus, the **magnetic field at P due to entire circular loop** is

$$\vec{B} = B_x \hat{i} = \int_0^{2\pi R} \frac{\mu_0}{4\pi} \frac{IRdl}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{IR \times 2\pi R}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i} \text{ Or } \boxed{\vec{B} = \frac{\mu_0}{2} \frac{IR^2}{(x^2 + R^2)^{\frac{3}{2}}} \hat{i}}$$

Applications of Biot-Savart Law

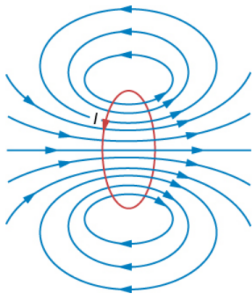


Figure 12.12 Sketch of the magnetic field lines of a circular current loop.

Magnetic Field On The Axis Of a Circular Current Loop

- **SPECIAL CASE:** The field at the center of the loop ($x = 0$).

$$\vec{B} = \frac{\mu_0}{2} \frac{IR^2}{(0 + R^2)^{\frac{3}{2}}} \hat{i}$$

$$\boxed{\vec{B} = \frac{\mu_0}{2} \frac{I}{R} \hat{i}}$$

- The magnetic field lines due to a circular wire form closed loops

Example 4.6

A straight wire carrying a **current of 12 A** is bent into a semi-circular arc of **radius 2.0 cm** as shown in Fig. 4.13(a). Consider the **magnetic field B** at the **center** of the arc.

- (a) What is the **magnetic field due to the straight segments**?
- (b) In what way the **contribution to B** from the **semicircle differs from that of a circular loop** and in **what way does it resemble**?
- (c) Would your answer be different **if the wire were bent** into a semi-circular arc of the same radius but **in the opposite way** as shown in Fig. 4.13(b)?

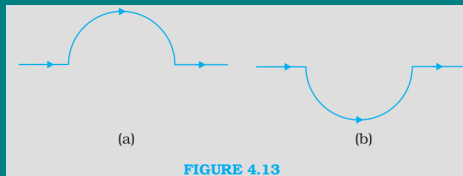
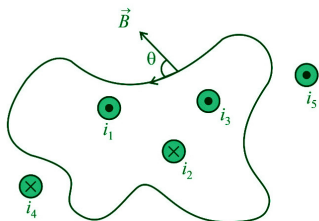


FIGURE 4.13

Example 4.7

Consider a tightly wound **100 turn** coil of **radius 10 cm**, carrying a **current of 1 A**.
What is the magnitude of the magnetic field at the center of the coil?

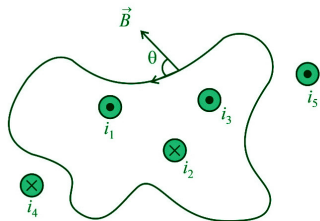
Ampere's Circuital Law



- Ampere's circuital law considers an **open surface with a boundary**.
- The **surface has current passing through it**.
- We consider the **boundary to be made up** of a number of **small line elements** of length dl .
- **Statement:** The **line integral** of the resultant magnetic field $\vec{B}$ along a closed curve is **equal to** μ_0 times the **algebraic sum of the currents** crossing the area bounded by the closed curve.

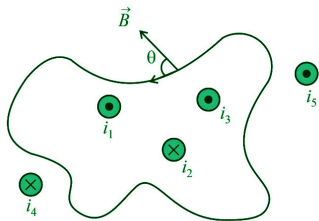
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

Ampere's Circuital Law



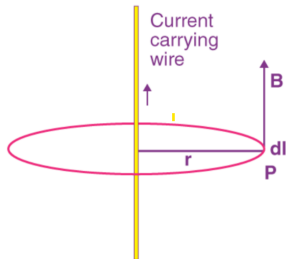
- Figure shows a plane **closed curve enclosing the currents** i_1, i_2, i_3 .
- **Before assigning the sign of the current**, assign a sense to the curve by **putting an arrow on the curve** which will give the **direction of $\vec{dl}$** for evaluating $\oint \vec{B} \cdot \vec{dl}$.
- Now, **curl the finger's of your right hand around the curve along the arrow**. The **stretched thumb gives the direction of the positive current**.
- Thus, i_1, i_3 are **positive** and i_2 is **negative**.

Ampere's Circuital Law



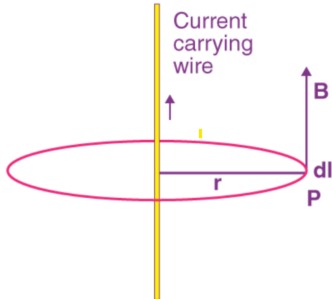
- Hence, $I_{enc} = i_1 - i_2 + i_3$
- The magnetic field $\vec{B}$ in $\oint \vec{B} \cdot d\vec{l}$ is the **net field** due to **all the currents enclosed or not enclosed by the closed curve**.
- The **result does not depend on the shape of the curve or position of the current carrying conductor** inside it.
- If $\oint \vec{B} \cdot d\vec{l} = 0$, it **does not necessarily mean** the $\vec{B} = 0$ everywhere in along the path. It **only mean** that **total current** through the area **bounded by the path is zero**.

Magnetic field due to straight infinite current-carrying wire



- For calculating the magnetic field around the wire we draw an **imaginary circle** of **radius r** from the wire.
- The **magnetic field does not vary with the change in the distance r** . The **length of the path** covered is $2\pi r$. Now using Ampere's Law, we find that the **magnetic field at a distance r** outside the wire is **tangential** and given by,

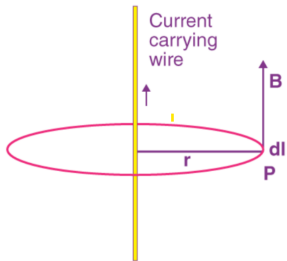
Magnetic field due to straight infinite current-carrying wire



$$\oint_0^{2\pi r} \vec{B} \cdot d\vec{l} = \mu_0 I$$
$$B \int_0^{2\pi r} dl = \mu_0 I$$
$$B \times 2\pi r = \mu_0 I$$

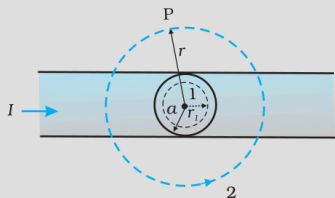
$$B = \frac{\mu_0 I}{2\pi r}$$

Magnetic field due to straight infinite current-carrying wire



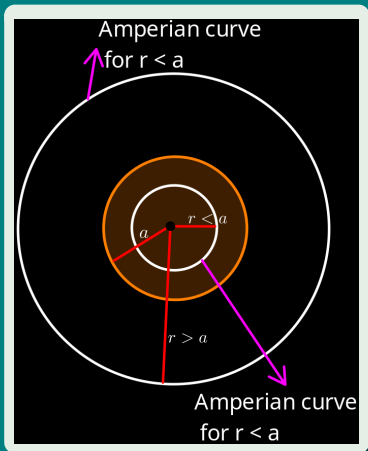
Some important Results:

- The magnetic field at every point on a circle of radius r , (with the wire along the axis), is same in magnitude (cylindrical symmetry).
- The **field direction** at any point on this circle is **tangential** to it.
- Magnetic **field lines form concentric circles or closed loop**. unlike the electrostatic field lines which originate from positive charges and end at negative charges.



Example 4.8

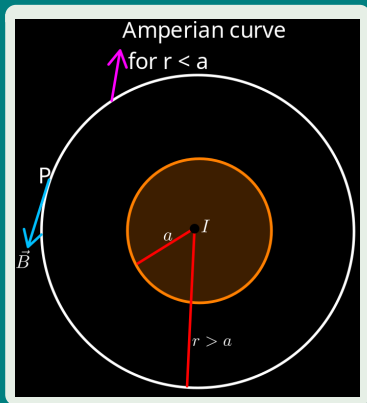
Figure shows a long straight wire of a circular cross-section (radius a) carrying steady current I . The current I is uniformly distributed across this cross-section. Calculate the magnetic field in the region $r < a$ and $r > a$.



Solution:

Let us draw Amperian curves as circles of radius r with the center at the axis of the cylindrical conductor. By symmetry, the magnetic field $\vec{B}$ at each point of a circle will be tangential to it with the same magnitude, So,

$$\oint_0^{2\pi r} \vec{B} \cdot d\vec{l} = B \times 2\pi r$$



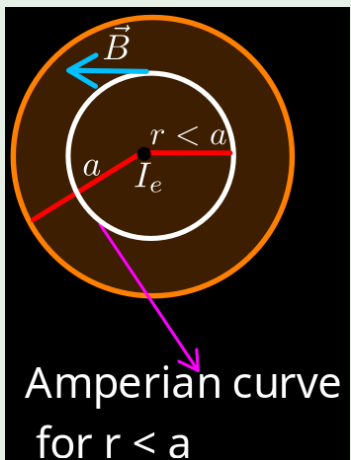
Solution:

Case 1: When $r > a$, the current enclosed within the Amperian curve is I . Hence

$$B \times 2\pi r = \mu_0 I \quad \text{Or}$$

$$B = \frac{\mu_0 I}{2\pi r} \quad \dots \text{for } r > a$$

Note: $B \propto \frac{1}{r}$



Solution:

Case 2: When $r < a$, the current enclosed within the Amperian curve is I_e . Hence

$$I_e = \frac{I}{4\pi a^2} \times 4\pi r^2,$$

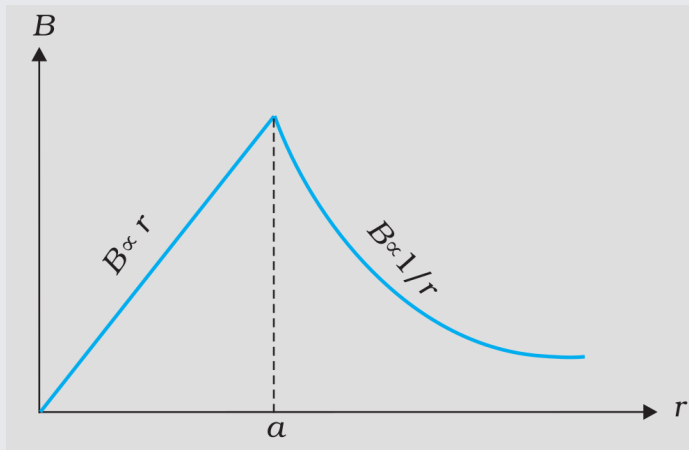
$$\text{Or, } I_e = \frac{I r^2}{a^2}$$

$$\text{Now, } B \times 2\pi r = \mu_0 I_e = \mu_0 \frac{I r^2}{a^2}$$

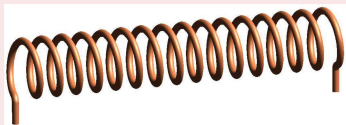
$$\boxed{B = \frac{\mu_0 I r}{2\pi a^2}} \dots \text{for } r < a$$

Note: $B \propto r$

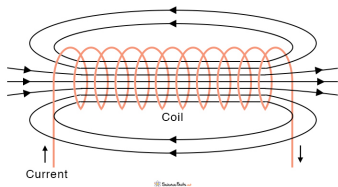
Graph of B versus r



Solenoids

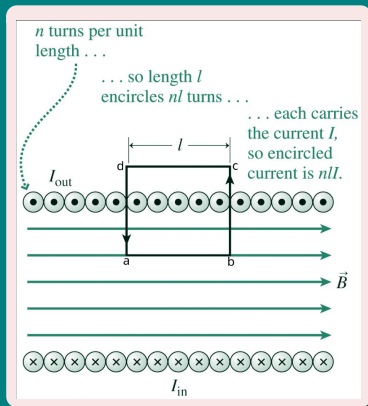


Solenoid Magnetic Field



- A **long wire wound** in the form of a **helical coil** is known as a **solenoid**.
- The **length of an ideal solenoid is large** compared to its cross-sectional dimension (radius).
- The **magnetic field inside an ideal solenoid** (very tightly wound, long) is **uniform and parallel to the solenoid axis** everywhere and is **zero outside** it, provided the field point is not near the ends of the solenoid.

Solenoids

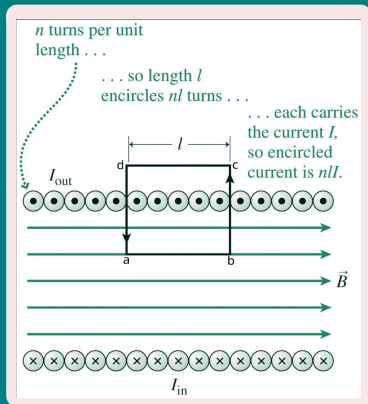


Magnetic field inside an ideal solenoid

- To calculate the magnetic field inside the solenoid, we draw a rectangle $abcd$ as shown in figure.
- Along segment ab , $\vec{B}$ is **uniform** and **parallel** to the path. Along segments bc and da , B is **perpendicular** to part of the path and vanishes over the rest of it. Along segment cd , $B = 0$.
- So the line integral of $\vec{B}$ along $abcd$ is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_e$$

Solenoids

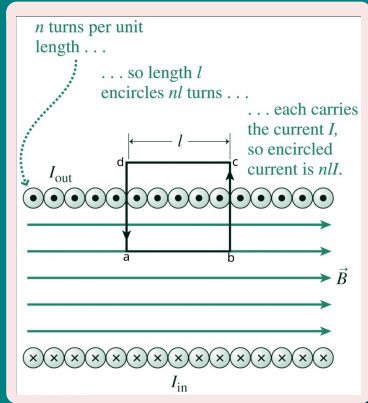


Magnetic field inside an ideal solenoid

$$\int_a^b \vec{B} \cdot d\vec{l} = \mu_0 I_e$$

- if n is the **number of turn per unit length**.
- The **total number of turns in the length l** is nl .
- **Each turn carries a current I** , so the **net current crossing $abcd$** is ($I_e = nIl$)

Solenoids



Magnetic field inside an ideal solenoid

$$\begin{aligned} B \int_a^b dl &= \mu_0 n I l \\ \vec{B} l &= \mu_0 n I l \\ \boxed{B} &= \mu_0 n I \end{aligned}$$

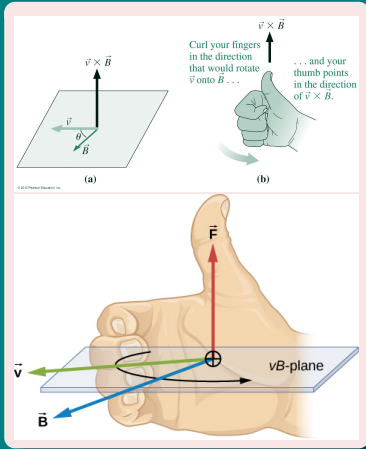
Lorentz Force

- Consider a **point charge** q moving with a **velocity** $\vec{v}$ in the **presence of** both **electric field** $\vec{E}$ and **magnetic field** $\vec{B}$
- The moving **charge experience two force**:
 - Electric Force**: $\vec{F}_e = q\vec{E}$ and,
 - Magnetic Force**: $\vec{F}_m = q\vec{v} \times \vec{B}$
- The **net force** on the moving charge is called **Lorentz force** F_L and is given by

$$\vec{F}_L = \vec{F}_e + \vec{F}_m$$
$$\vec{F}_L = q\vec{E} + q\vec{v} \times \vec{B}$$

$$\boxed{\vec{F}_L = q \left(\vec{E} + \vec{v} \times \vec{B} \right)}$$

Lorentz Force



- **Force due to magnetic field depends on q , $\vec{v}$ and $\vec{B}$.**
- Value of **magnetic force** will be **zero** if
 - ① **Charge is at rest $v = 0$**
 - ② **$\vec{v}$ is parallel to $\vec{B}$ (i.e., $\sin(\theta) = 0$)**
 - ③ **$q = 0$ Or $B = 0$**
- **Magnetic Lorentz force $\vec{F}_m$ will always perpendicular to both $\vec{v}$ and $\vec{B}$ and its direction is given by **Right hand thumb rule** for vector product.**

Lorentz Force

- **Magnetic Lorentz force help us to define the unit of magnetic field**

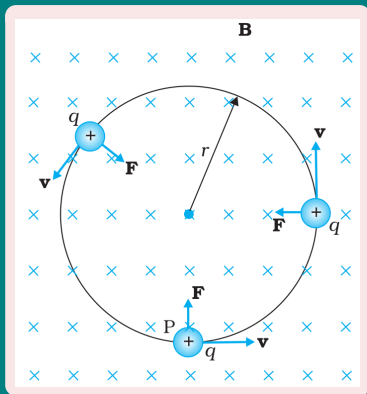
$$B = \frac{F}{qv \sin \theta}$$

The **S.I unit of magnetic field is Tesla (T)**

$$1T = \frac{N \ s}{C \ m}$$

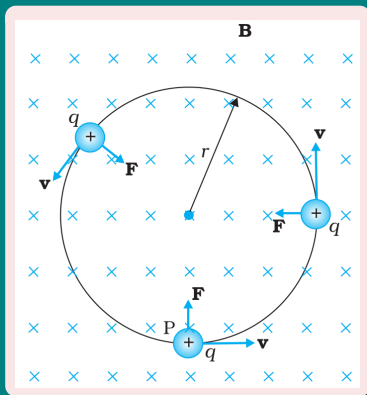
- A **smaller unit**, called the **gauss (G)**, where $1G=10^{-4} \ T$, is sometimes used.

Motion of a Charged Particle in a Magnetic Field



- Consider the **motion of a charge particle** in **uniform magnetic field** $\vec{B}$
- The magnetic **force is always perpendicular to velocity**, so that it does **no work** on the charged particle ($dW = \vec{F} \cdot d\vec{r} = 0$).
- The particle's **kinetic energy and speed thus remain constant**.
- The **direction of motion is affected but not the speed**.

Motion of a Charged Particle in a Magnetic Field



Case 1: When $\vec{v}$ perpendicular to $\vec{B}$

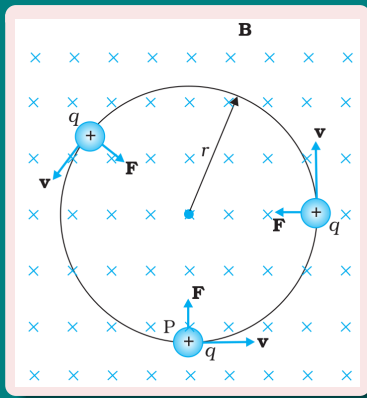
- Force acting on charge particle (q) is:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

- This magnetic force qvB acts as **centripetal force** $\frac{mv^2}{r}$ and produces a circular motion perpendicular to the magnetic field.
- Here, r be the radius of the circle, then

$$\frac{mv^2}{r} = qvB \text{ Or } \boxed{r = \frac{mv}{qB}}$$

Motion of a Charged Particle in a Magnetic Field



Case 1: When $\vec{v}$ perpendicular to $\vec{B}$

- The time taken for one revolution is

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

- If ω is the angular frequency, then

$$\omega = \frac{2\pi}{T} = \frac{qB}{m}$$

Motion of a Charged Particle in a Magnetic Field

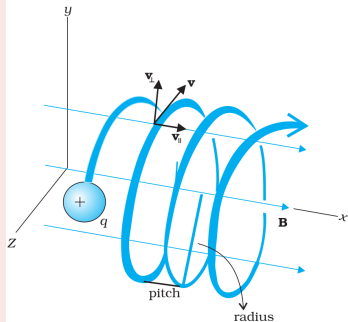


FIGURE 4.6 Helical motion

Case 2: If velocity $\vec{v}$ has a component along $\vec{B}$

- In this case the **component of velocity along $\vec{B}$** (i.e., $v_{\parallel}$) make the particle move along the field and the **path of the particle** would be a **helical**.
- The **distance moved** along the magnetic field in **one rotation** is called **pitch p**.

$$p = v_{\parallel} \times T = v \cos \theta \times \frac{2\pi m}{qB}$$

$$p = \frac{2\pi m v \cos \theta}{qB}$$

Example 4.3

What is the radius of the path of an electron (mass $9 \times 10^{-31} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$) moving at a speed of $3 \times 10^7 \text{ m/s}$ in a magnetic field of $6 \times 10^{-4} \text{ T}$ perpendicular to it? What is its frequency? Calculate its energy in keV. ($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$).

Exercise 4.11

In a chamber, a uniform magnetic field of 6.5 G ($1 \text{ G} = 10^{-4} \text{ T}$) is maintained. An electron is shot into the field with a speed of $4.8 \times 10^6 \text{ ms}^{-1}$ normal to the field. Explain why the path of the electron is a circle. Determine the radius of the circular orbit. ($e = 1.5 \times 10^{-19} \text{ C}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$)

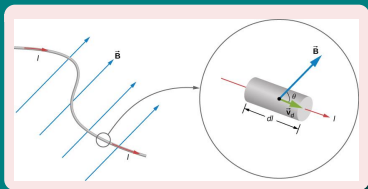
Exercise 4.12

In Exercise 4.11 obtain the frequency of revolution of the electron in its circular orbit. Does the answer depend on the speed of the electron? Explain.

Reference Example

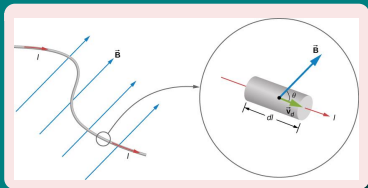
A proton enters a uniform magnetic field of $1.0 \times 10^{-4} \text{ T}$ with a speed of $5 \times 10^5 \text{ m/s}$. At what angle must the magnetic field be from the velocity so that the pitch of the resulting helical motion is equal to the radius of the helix?

Magnetic force on a current-carrying conductor



- Electric current is **flow of charge**. A **current - carrying wire** in a **magnetic field** ($\vec{B}$) must therefore **experience a force** due to the field.
- let's consider the **infinitesimal section of wire** of **length** dl and **area of cross section** A .
- So its **volume** is $V = Adl$.
- Let, n be the **charge carriers per unit volume**.
- So, the **number of charge carriers in the section** is $nV = nAdl$
- If the **charge carriers** move with **drift velocity** $\vec{v}_d$

Magnetic force on a current-carrying conductor



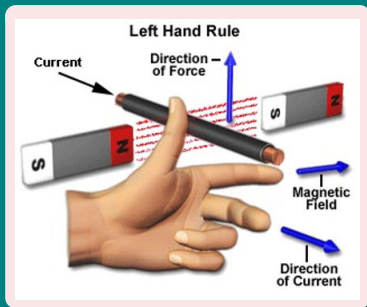
- **Force on the charge carriers** moving with velocity $\vec{v}_d$ in the section of wire is

$$d\vec{F} = Q\vec{v}_d \times \vec{B}$$
$$d\vec{F} = neA dl \vec{v}_d \times \vec{B}$$

- We can define $d\vec{l}$ to be a vector of length dl pointing along $\vec{v}_d$ (or I), which allows us to rewrite this equation as

$$d\vec{F} = neAv_d d\vec{l} \times \vec{B}$$
$$\boxed{d\vec{F} = I d\vec{l} \times \vec{B}} \quad (\because I = neAv_d)$$

Magnetic force on a current-carrying conductor



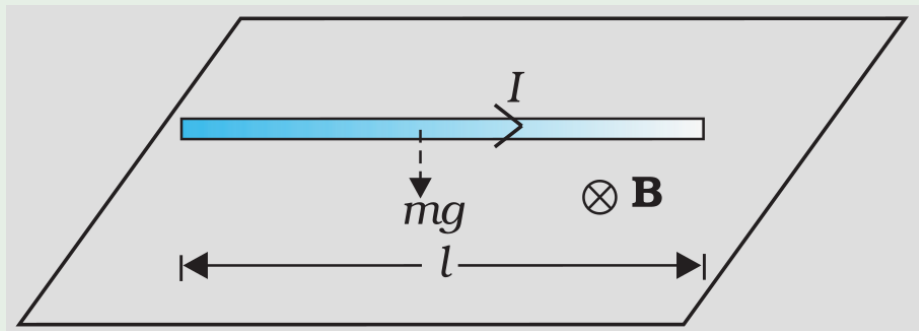
- The force on a straight, current-carrying wire in a uniform magnetic field.

$$\vec{F} = I\vec{l} \times \vec{B}$$

- The direction of this force is given by Fleming's Left hand rule.

Example 4.1

A straight wire of mass 200 g and length 1.5 m carries a current of 2 A . It is suspended in mid-air by a uniform horizontal magnetic field B . What is the magnitude of the magnetic field?



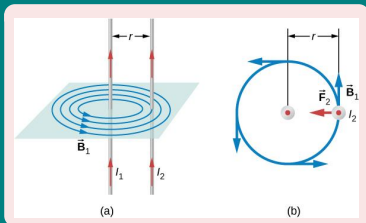
Exercise 4.5

What is the magnitude of magnetic force per unit length on a wire carrying a current of 8 A and making an angle of 30° with the direction of a uniform magnetic field of 0.15 T ?

Exercise 4.6

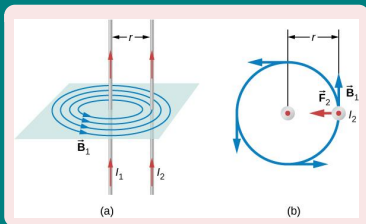
A 3.0 cm wire carrying a current of 10 A is placed inside a solenoid perpendicular to its axis. The magnetic field inside the solenoid is given to be 0.27 T. What is the magnetic force on the wire?

FORCE BETWEEN TWO PARALLEL CURRENTS



- We know that a **magnetic field** is generated by a **current-carrying conductor**.
- Another **current-carrying conductor** experiences **force** as a result of the **external magnetic field**.
- Consider **two long parallel current-carrying wires**, separated by a **distance 'r'**, such that **one of the wires** is carrying current I_1 and the **other** is carrying I_2 .

FORCE BETWEEN TWO PARALLEL CURRENTS



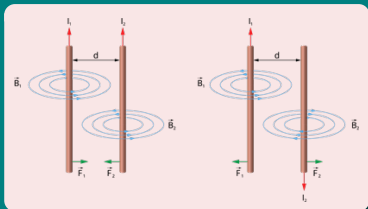
- The **magnitude of the field due to I_1** at a **distance r** is

$$B_1 = \frac{\mu_0 I_1}{2\pi r} \dots\dots(1)$$

- This **field (B_1)** produced by wire 1 **exert a force (F_{21}) on a length l of wire 2** and is given by.

$$F_{21} = I_2 l B_1$$
$$F_{21} = \frac{\mu_0 I_1 I_2 l}{2\pi r} \dots\dots(2)$$

FORCE BETWEEN TWO PARALLEL CURRENTS



- Similarly, **field** (B_2) produced by wire 2 **exert a force** (F_{12}) **on a length** l **of wire 1** and its direction is opposite to F_{21}

$$F_{21} = I_1 l B_2$$
$$F_{12} = \frac{\mu_0 I_1 I_2 l}{2\pi r} \dots\dots (3)$$

$$\vec{F}_{12} = - \vec{F}_{21} \text{ (consistent with Newton's third law)}$$

- **Parallel currents (same direction) attract, and antiparallel currents (opposite direction) repel.**

FORCE BETWEEN TWO PARALLEL CURRENTS

Definition of S.I. Unit of current, Ampere

- The **definition of the ampere** is based on the force between current - carrying wires. Note that for long, parallel wires separated by **1 meter** with each carrying **1 ampere**, the **force per meter** is

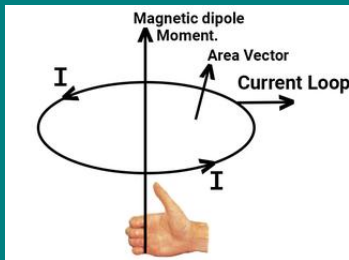
$$\frac{F}{l} = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{r}$$

$$\frac{F}{l} = \left(10^{-7} \frac{T \cdot m}{A}\right) \frac{2 A^2}{1 m} = 2 \times 10^{-7} N/m$$

Exercise 4.7

Two long and parallel straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire A.

Circular current loop as a magnetic dipole



- When a **current passes through a circular loop** it is **considered a magnetic dipole**.
- The **magnetic dipole moment $\vec{m}$** of a current loop is **defined as the product of the current (I) in the loop and area of the loop (A)**.

$$\vec{m} = N I \vec{A}$$

Here, **N is the no. of turns** in the loop

- The **direction of magnetic dipole is determine** according to **right hand thumb rule**.

Circular current loop as a magnetic dipole

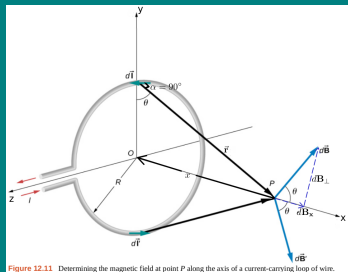


Figure 12.11 Determining the magnetic field at point P along the axis of a current-carrying loop of wire.

- **Magnetic field on the axis of a circular current loop of radius R and carrying steady current (I) is**

$$B = \frac{\mu_o I R^2}{2(x^2 + R^2)^{3/2}}$$

for $x \gg R$, ($x^2 + R^2 \approx x^2$)

$$B = \frac{\mu_o I R^2}{2x^3}$$

$$B = \frac{\mu_o I R^2}{2\pi x^3} \frac{2\pi}{2\pi} = \frac{2\mu_o I A}{4\pi x^3} (\because A = \pi R^2)$$

$$B = \frac{\mu_o 2m}{4\pi x^3} (\because m = IA)$$

Circular current loop as a magnetic dipole

- This **above expression** is **similar** to **electric field due to an electric dipole on its axis.**

$$E = \frac{2p}{4\pi\epsilon_0 x^3}$$

- Similarly, **Electric field** on the **perpendicular bisector of the electric dipole** is give by.

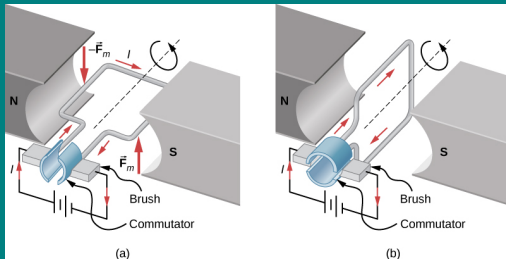
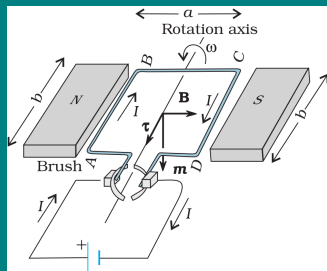
$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{x^3}$$

Circular current loop as a magnetic dipole

- If we replace p by m , $\frac{1}{\epsilon_o}$ by μ_o and E by B , We obtain the result for magnetic field B for point on the plane of the current loop.

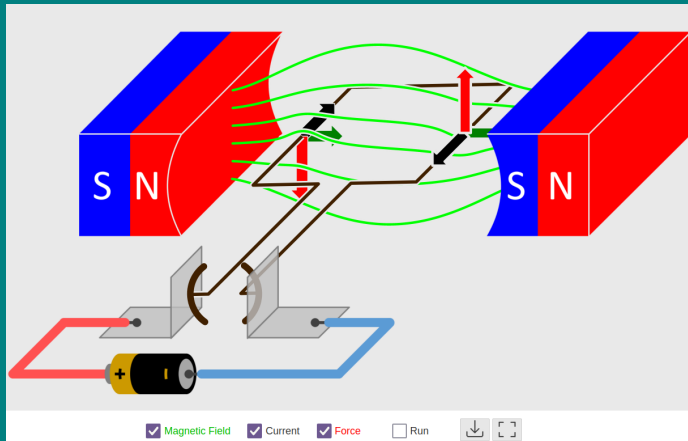
$$B = \frac{\mu_o}{4\pi} \frac{m}{x^3}$$

Torque on a Current Loop In a Uniform Magnetic Field

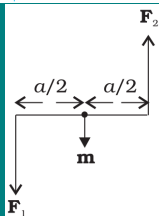
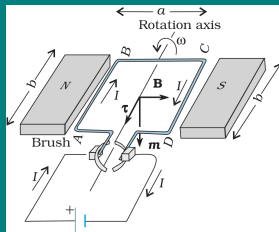


- A rectangular loop carrying a steady current I and placed in a uniform magnetic field experiences a torque. It does not experience a net force.
- This behaviour is **analogous to** that of electric **dipole** in a uniform electric field.

(1) When uniform magnetic field B is in the plane of the loop



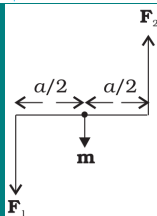
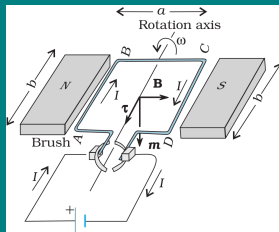
Torque on a Current Loop In a Uniform Magnetic Field



Case 1: When uniform magnetic field $\mathbf{B}$ is in the plane of the loop

- The field exerts **no force** on the two **arms AD and BC** of the loop ($\because I$ and $\mathbf{B}$ are in the same direction).
- The force F_1 on side AB is
$$\boxed{F_1 = IbB}$$
(into the plane of loop)
- Similarly, force F_2 on the arm CD
$$\boxed{F_2 = IbB}$$
(out of the plane of loop)
- Thus, the **net force on the loop is zero**.

Torque on a Current Loop In a Uniform Magnetic Field



Case 1: When uniform magnetic field B is in the plane of the loop

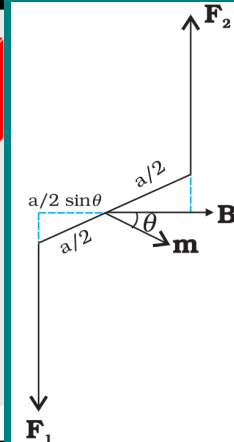
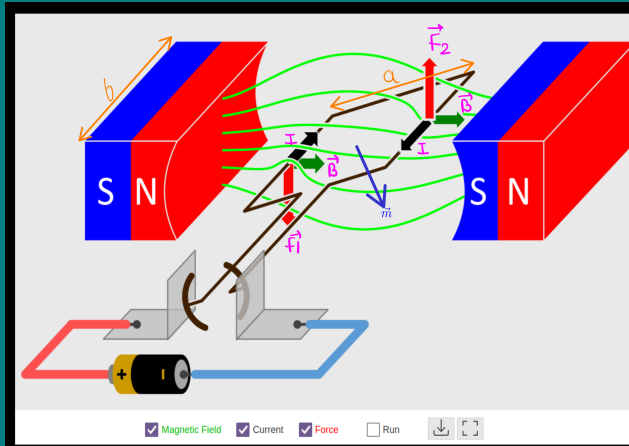
- There is a **torque on the loop** due to the pair of **forces F_1 and F_2** .
- Magnitude of this torque is

$$\tau = F_1 \frac{a}{2} + F_2 \frac{a}{2}$$

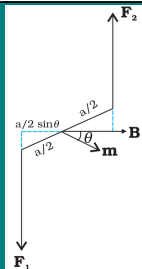
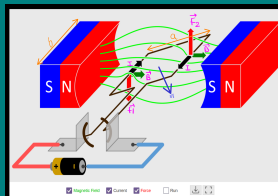
$$\tau = IbB \frac{a}{2} + IbB \frac{a}{2} = IabB$$

$$\boxed{\tau = IAB} \text{ (Here, } A = ab \text{ is the area of the loop)}$$

(2) When the plane of the loop, is not along the magnetic field



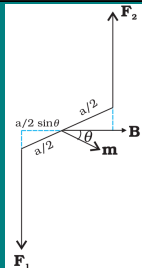
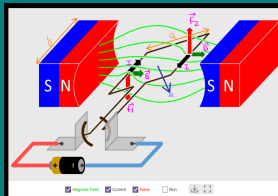
Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

- Let the **angle between the field and the normal** to the coil to be angle θ
- The **forces on the arms BC and DA are equal, opposite, and act along the axis** of the coil. Being **collinear** along the axis they cancel each other, **resulting in no net force or torque.**

Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

- The forces on arms AB and CD are F_1 and F_2 . They too are **equal and opposite**, with magnitude,

$$F_1 = F_2 = I b B$$

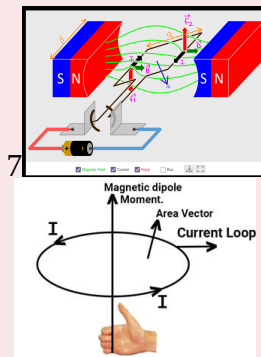
- But they are **not collinear**! This **results in a couple** as before. The torque on the loop is,

$$\tau = F_1 \frac{a}{2} \sin \theta + F_2 \frac{a}{2} \sin \theta$$

$$\tau = \frac{a \sin \theta}{2} (F_1 + F_2)$$

Torque on a Current Loop In a Uniform Magnetic Field

Case 2: When the plane of the loop, is not along the magnetic field



$$\tau = \frac{a \sin \theta}{2} (2IbB) = IabB \sin \theta$$

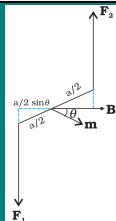
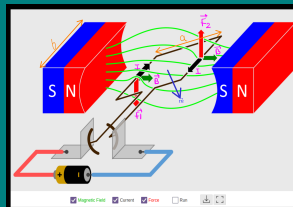
$$\tau = IAB \sin \theta \quad (\text{Here, } A \text{ is the area of the loop})$$

- Now, we **define** a quantity **magnetic moment** ($\vec{m}$) of the current loop as,

$$\vec{m} = I\vec{A}$$

- The **direction** of the **area vector** $\vec{A}$ is **given by the right-hand thumb rule**.

Torque on a Current Loop In a Uniform Magnetic Field



Case 2: When the plane of the loop, is not along the magnetic field

- This **torque** ($\vec{\tau}$) can be **expressed** as **vector product** of the **magnetic moment** ($\vec{m}$) of the coil **and** the **magnetic field** ($\vec{B}$)

$$\vec{\tau} = \vec{m} \times \vec{B}$$

- This is **analogous to** the **Electric dipole** of **dipole moment** $\vec{p}$ in an **electric field** $\vec{E}$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

Torque on a Current Loop In a Uniform Magnetic Field

Case 2: When the plane of the loop, is not along the magnetic field

- The **torque** ($\vec{\tau}=0$) **vanishes** when $\vec{m}$ is either **parallel or antiparallel** to the **magnetic field** $\vec{B}$
- When $\vec{m}$ and $\vec{B}$ are **parallel** ($\theta = 0^\circ$) the **equilibrium is a stable one**. Any **small rotation** of the coil **produces a torque** which **brings it back** to its **original position**.
- When they are **antiparallel** ($\theta = 180^\circ$), the **equilibrium is unstable**.
- If the **loop has N** closely wound **turns**, the expression for **torque** is

$$\vec{\tau} = N I \vec{A} \times \vec{B}$$

Example 4.11

A **100 turn** closely wound circular coil of **radius 10 cm** carries a **current of 3.2 A**.

(a) **What is the field at the centre of the coil?**

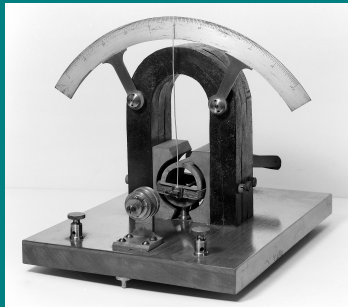
(b) **What is the magnetic moment of this coil?**

The coil is placed in a vertical plane and is free to rotate about a horizontal axis which coincides with its diameter. A **uniform magnetic field of 2 T** in the horizontal direction exists such that **initially the axis of the coil is in the direction of the field**. The coil **rotates** through an **angle of 90°** under the influence of the magnetic field.

(c) **What are the magnitudes of the torques on the coil in the initial and final position?**

(d) **What is the angular speed** acquired by the coil when it has rotated by 90° ? The **moment of inertia** of the coil is 0.1 kg m^2 .

THE MOVING COIL GALVANOMETER



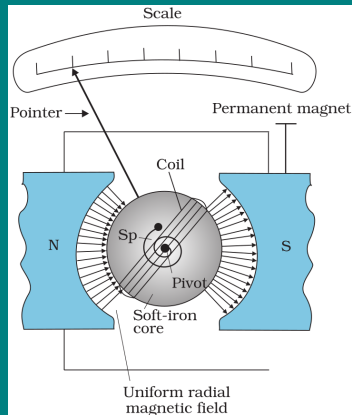
Moving Coil Galvanometer (MCG)

- A **moving coil galvanometer** is an instrument which is **used to detect and measure electric currents**.

Working Principle of A MCG

- A **current-carrying coil**, when **placed in an external magnetic field**, **experiences a torque**, which **deflects the coil**. and the **deflection is directly proportional to the amount of the current** passed through the coil.

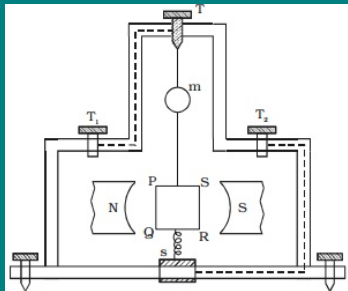
THE MOVING COIL GALVANOMETER



Construction of A (MCG)

- The **galvanometer consists of a coil, with many turns (N), free to rotate** about a fixed axis in a **uniform radial magnetic field B**
- There is a **cylindrical soft iron core** which not only **makes the field radial** but **also increases the strength of the magnetic field.**
- The **lower end of the coil is connected to a spring (S_P)**

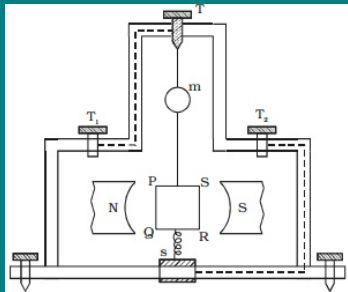
THE MOVING COIL GALVANOMETER



Working of A (MCG)

- When a current (I) flows through the coil, a **torque acts on it**. This torque is given by $\tau = NIAB$ ($\sin \theta = 1$ since the field is radial.)
- This **magnetic torque ($NIAB$) tends to rotate the coil**.
- The **spring provides a counter torque ($k\phi$) that balances the magnetic torque ($NIAB$); resulting in a steady angular deflection (ϕ).**

THE MOVING COIL GALVANOMETER



Working of A (MCG)

- In equilibrium

$$k\phi = NIAB$$

$$\phi = \frac{NIAB}{k}$$

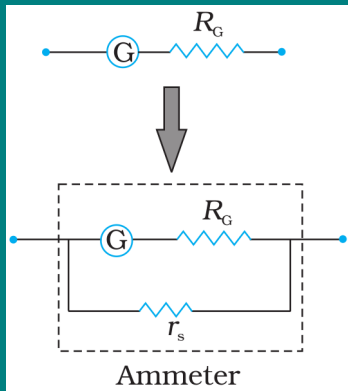
- where, k is the **torsion constant of the spring**.
- i.e $I \propto \phi$. Since the **deflection is directly proportional to the current** flowing through the coil, the **scale is linear** and is **calibrated to give directly the value of the current**.

THE MOVING COIL GALVANOMETER

Conversion of Galvanometer into an ammeter

- The **galvanometer cannot as such be used as an ammeter to measure the value of the current** in a given circuit. This is **for two reasons**:
- **(i) Galvanometer is a very sensitive device, it gives a full-scale deflection for very small currents of the order of μA . A large current cannot be passed through the galvanometer, as it may damage the coil.**
- **(ii) For measuring currents, the galvanometer has to be connected in series, and as it has a large resistance, this will change the value of the current in the circuit.**

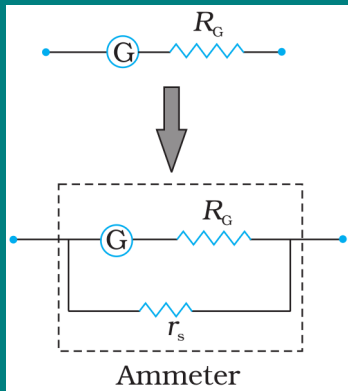
THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an ammeter

- However, a **galvanometer is converted into an ammeter by connecting a small resistance (r_s), called shunt resistance, in parallel with galvanometer.**
- As a result, **when large current flows in a circuit, only a small fraction of the current passes through the galvanometer and the remaining larger portion of the current passes through the small shunt resistance.**

THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an ammeter

- The **effective resistance of the ammeter** R_a is

$$\frac{1}{R_a} = \frac{1}{R_G} + \frac{1}{r_s} = \frac{r_s + R_G}{r_s R_G}$$

$$R_a = \frac{r_s R_G}{r_s + R_G} \approx r_s \quad (\text{if } R_G \gg r_s)$$

- If r_s has **small** value, the **effect of** introducing the measuring instrument is also **small and negligible**.

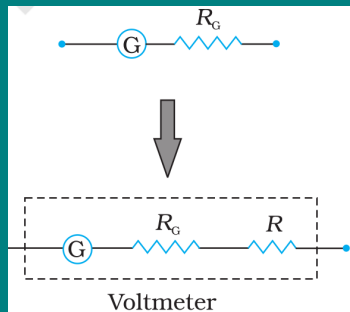
THE MOVING COIL GALVANOMETER

Current sensitivity of a Galvanometer

- The **current sensitivity** of the **galvanometer** is **defined** as the **deflection** (ϕ) **per unit current** (I).

$$\frac{\phi}{I} = \frac{NAB}{k}$$

THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an voltmeter

- **Voltmeter** is an **instrument used to measure potential difference** between the two ends of a current carrying conductor. it must be **connected in parallel**.
- A **galvanometer** can be **converted into a voltmeter** by **connecting a high resistance (R) in series** with it.
- **Resistance of the voltmeter is**

$$R_v = R_G + R \approx R \text{ (Very large)}$$

THE MOVING COIL GALVANOMETER

Conversion of Galvanometer into an voltmeter

- R_v is **very large**, and hence a **voltmeter is connected in parallel** in a circuit as it **draws the least current from the circuit**. and the **measurement of potential difference is more accurate**.

Voltage sensitivity of a Galvanometer

- The **voltage sensitivity** of the **galvanometer** is **defined** as the **deflection (ϕ) per unit voltage (V)**.

$$\frac{\phi}{V} = \frac{NAB I}{kV} = \frac{NAB}{R} \left(\because \frac{I}{V} = \frac{1}{R} \right)$$

THE MOVING COIL GALVANOMETER

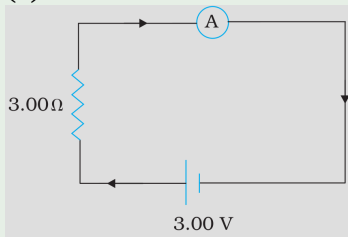
Voltage sensitivity of a Galvanometer

- **Note:** Increasing the current sensitivity may not necessarily increase the voltage sensitivity.
- The **current sensitivity** is given by : $\frac{\phi}{I} = \frac{NAB}{k}$
- **If we double the turns $N \rightarrow 2N$, then current sensitivity will also doubles**
 $\frac{\phi}{I} \rightarrow 2\frac{\phi}{I}$.
- **But, By doubling the turns $N \rightarrow 2N$ the resistance of the coil also doubles $R \rightarrow 2R$, thus the voltage sensitivity $\frac{\phi}{V} \rightarrow \frac{\phi}{V}$ remains unchanged.**

Example 4.13

In the circuit (Fig. 4.27) the current is to be measured. What is the value of the current if the ammeter shown

- (a) is a galvanometer with a resistance $R_G = 60.00 \, \Omega$;
- (b) is a galvanometer described in (a) but converted to an ammeter by a shunt resistance $r_s = 0.02 \, \Omega$;
- (c) is an ideal ammeter with zero resistance?



Exercise 4.10

Two moving coil meters, M_1 and M_2 have the following particulars:

$$R_1 = 10 \, \Omega, N_1 = 30, A_1 = 3.6 \times 10^{-3} \, \text{m}^2, B_1 = 0.25 \, \text{T}$$

$$R_2 = 14 \, \Omega, N_2 = 42,$$

$$A_2 = 1.8 \times 10^{-3} \, \text{m}^2, B_2 = 0.50 \, \text{T}$$

(The spring constants are identical for the two meters).

Determine the ratio of (a) current sensitivity and (b) voltage sensitivity of M_2 and M_1 .